

Effect of Atomization Variables on Powder Characteristics in the High-Pressured Water Atomization Process

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High-pressured water atomization is a viable production method for fine metal powders used for metal injection moulding and for additives in sintered parts. The atomization mechanism using two types of water jet nozzle (Cone and V-jet nozzle) under high pressure up to 70MPa was studied.

The pulverizing capability under high-pressured water atomization strongly depends on a gas stream induced by the ejecting effect of water jets, which surround the melt stream. A mean particle diameter of less than 10 μm can be obtained by the V-jet nozzle.

Ferrous metal powders are commonly used in the manufacture of sintered mechanical components. All of the powders now tend to be produced by the water atomization process because of its high productivity. Meanwhile, the need to increase the strength of the sintered mechanical components has been growing, and partially prealloyed powders which are

produced by allowing metal or alloy fine powders adhere to ferrous powders (1) have been attracting the attention of the industry. As it is difficult to produce by the conventional method, high-pressured water atomization was tried for this purpose.

In addition, finer powders have been required for metal injection mouldings or paints. In

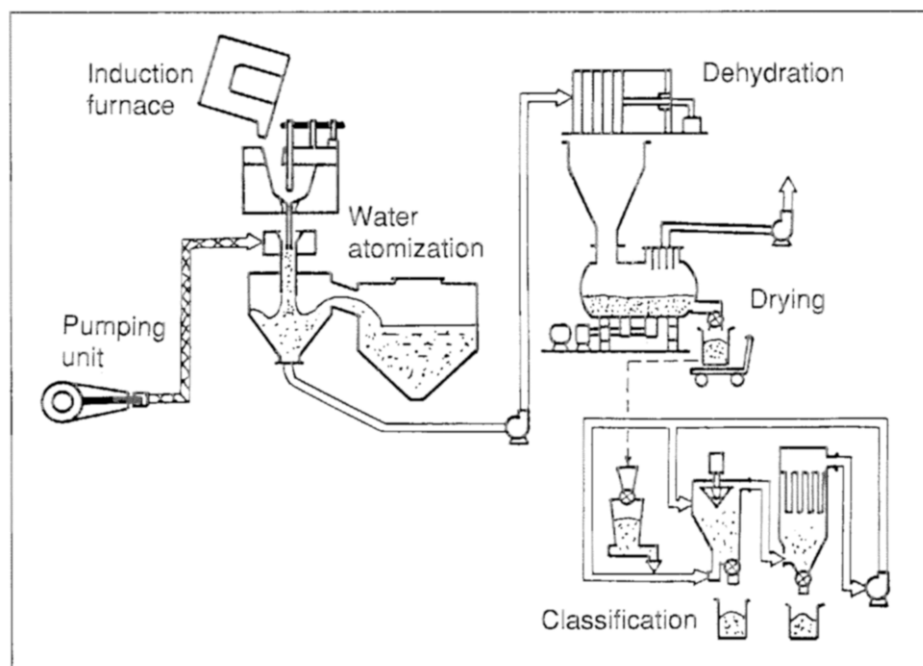


FIG. 1 Schematic diagram of high-pressured water atomization process.

particular, for metal injection mouldings, iron or nickel powders produced by the carbonyl process have often been used conventionally, but with increasing demands for high strength parts, high-pressured atomization which can produce high alloy fine powders is increasingly attracting the industry's attention.

POWDER PRODUCTION PROCESS

Fig. 1 shows schematically the high-pressure water atomization process used in this experiment. Raw materials are melted in a high-frequency melting furnace. The hot metal is charged into the tundish with a nozzle hole provided at the bottom, through which the hot metal flows down. A water jet pressurized up to 70MPa by high-pressure pumps sprays the hot metal stream to atomize it. The powders produced are dehydrated by a filter press, then heated and dried in a vacuum. The dried powders are classified by an air classifier according to the users' requirements.

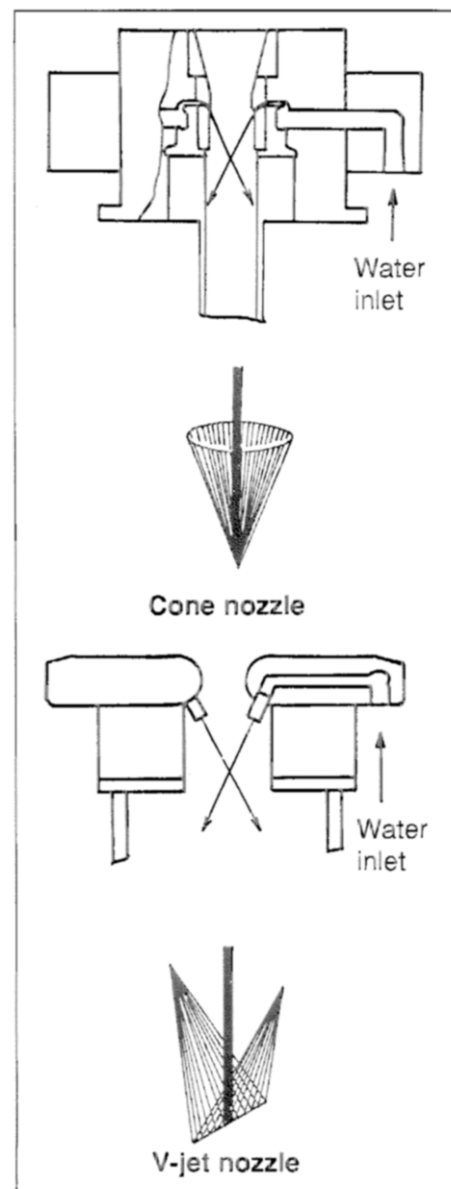


FIG. 2 Atomizing water jet nozzle and configurations.

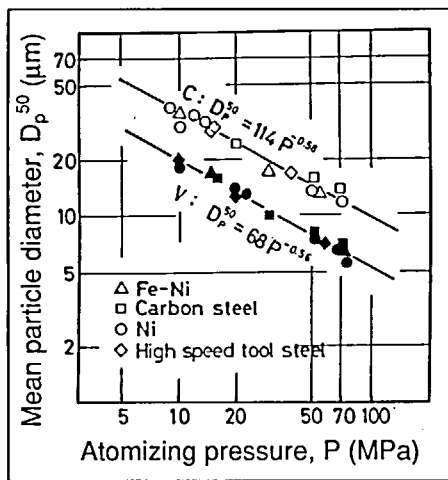


FIG. 3 Relationship between mean particle diameter and atomizing pressure.

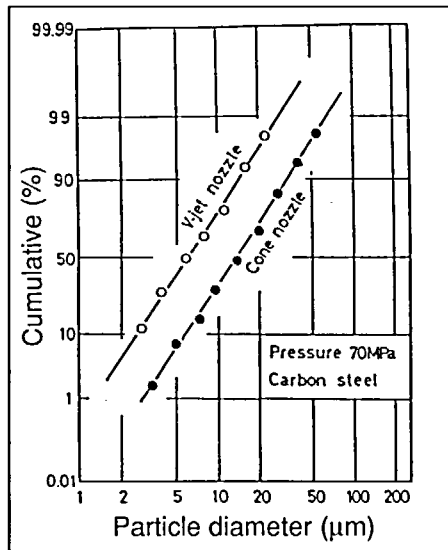


FIG. 4 Particle size distributions atomized by V-jet and Cone nozzles.

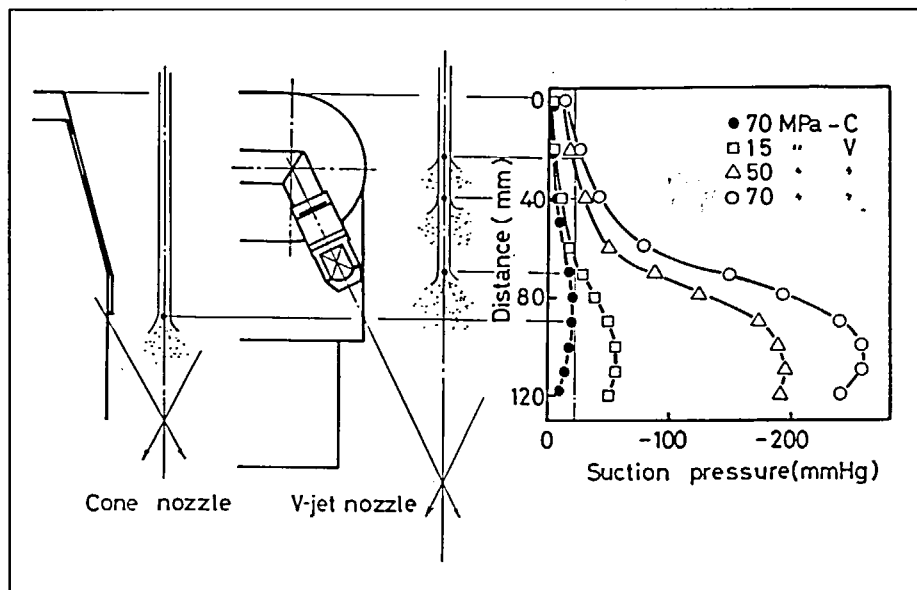


FIG. 5 Schematic view of disintegrating and pressure distributions sucked by water jet.

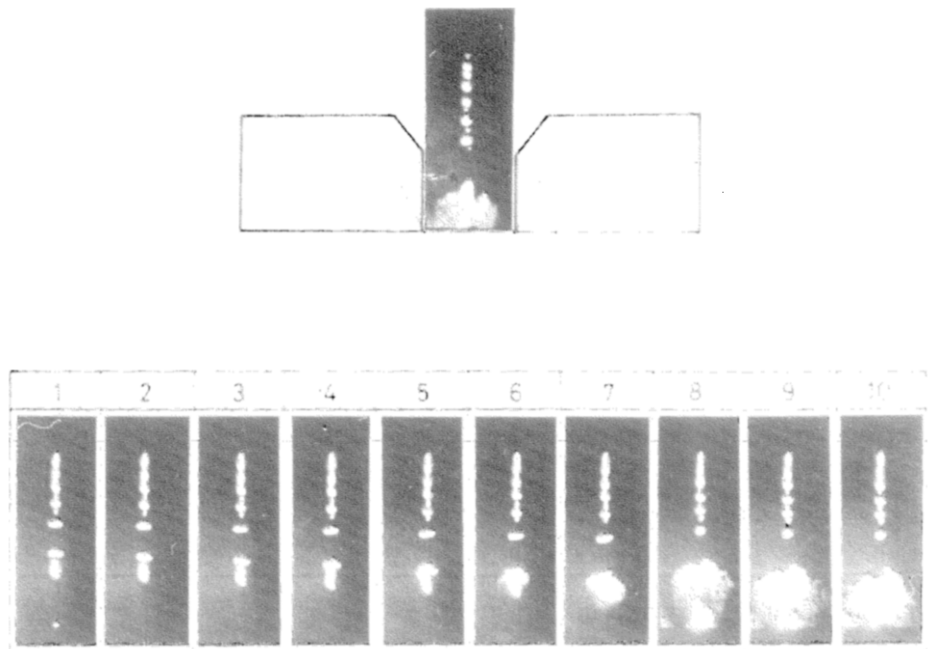


FIG. 6 Disintegrating behaviour shown by a high speed camera.

CHARACTERISTICS OF THE WATER JET NOZZLE USED

Fig. 2 shows two types of water jet nozzles used in this experiment. The nozzle shown in Fig. 2 (a) is the same type as that developed by Takeda (2), and spouts out high-pressured water jets in a cone form through an annular slit (Cone nozzle). Fig. 2 (b) shows a V-jet nozzle developed by Kobe Steel Ltd, in which multiple pencil-type nozzles are arranged.

Fig. 3 shows the relation between the mean particle diameter and the atomized pressure for each nozzle. The mean particle diameters atomized by both water jet nozzles show an almost similar dependence on pressure, but the V-jet nozzle shows a higher pulverizing capability than the Cone nozzle. Fine metal powders of a mean particle diameter of 7 μm

were obtained in the experiment. Fig. 4 shows particle size distributions of the produced powders, and it can be seen that there are no marked differences in the particle size distribution.

PRIMARY ATOMIZATION OF INFLOW GAS STREAM

Fig. 5 shows suction pressure distributions in the vertical direction when a water jet is spouted from a Cone nozzle or from V-jet nozzles. As shown in Fig. 5, the V-jet nozzle has a greater ejection effect than the Cone nozzle, providing a greater gas inflow rate. When using the V-jet nozzle, a space which is formed between adjoining jet curtains allows gas to flow more easily. In contrast, the Cone nozzle has no space between jet curtains, thus there is less of a gas inflow stream.

Fig. 6 shows the disintegrating behaviour of a hot metal stream. The photo shows that the hot metal stream has already been split by an inflow gas stream before it collides with the water jet. Fig. 5 also shows schematically the initial site of the split in the hot metal stream. The hot metal splits at the upper part of the jet intersection in both nozzles, but the V-jet nozzle has a higher splitting position.

To investigate the effects of the gas inflow rate, an inner sleeve was installed at the water jet passage, and experiments were conducted with the V-jet nozzle. Fig. 7 shows the relationship between the mean particle diameter of the produced powders and the suction pressure which is considered to be a parameter representing the gas inflow rate. With the decrease in suction pressure, that is, the gas inflow rate, the mean particle size of produced powders increases. Fig. 8 shows the relationship between the increase of the oxygen contents of powders and the suction pressure. With the decrease in the gas inflow rate, the oxygen content also decreases.

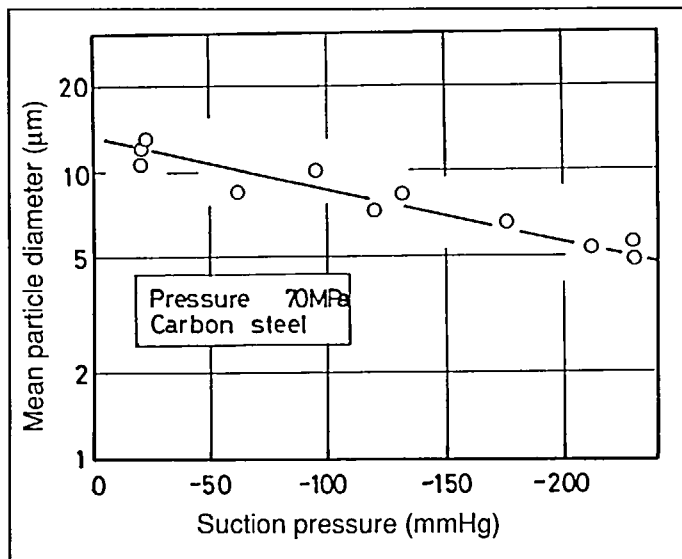


FIG. 7 Relationship between mean particle diameter and suction pressure

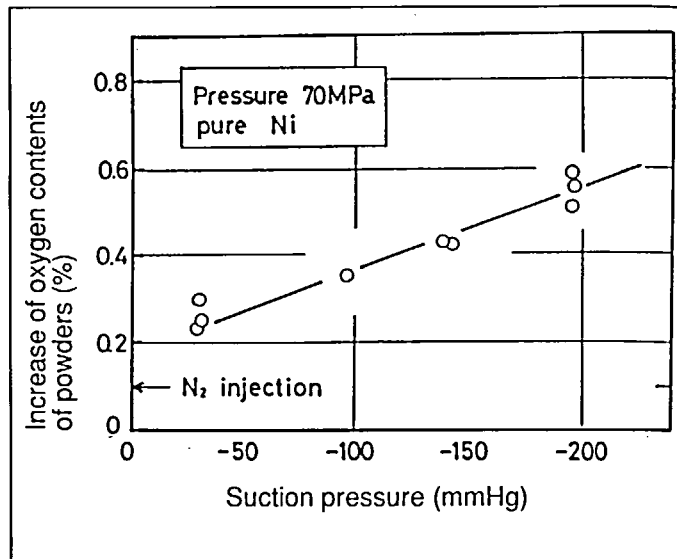


FIG. 8 Relationship between increase of oxygen contents and suction pressure

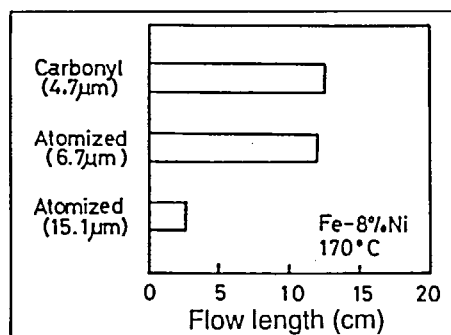


FIG. 9 Fluidity of mixed plastisols by spiral flow test.

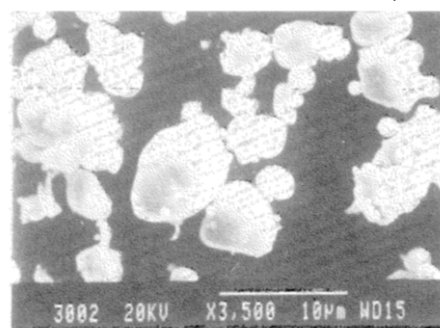
As a result, it is clear that the primary atomization of the hot metal stream by an inflow gas stream using a V-jet nozzle contributes greatly to reducing the particle size.

CHARACTERISATION OF PRODUCED POWDERS

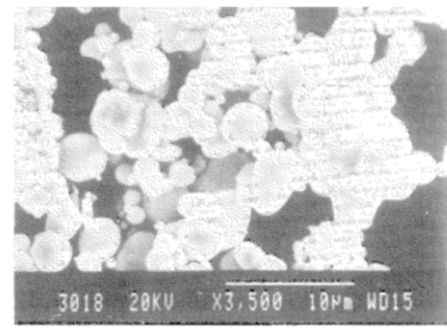
To investigate the applicability of high-pressured atomized powders to metal injection moulding, the flow properties of powders were examined by a spiral flow test. The results are shown in Fig. 9 and are compared with conventional carbonyl powders. When the mean particle diameter of the powders is coarse, the powders show a poor fluidity, but when the particle size becomes finer, it shows a fluidity equivalent to that of carbonyl powders. SEM micrographs of some of the alloy powders atomized are shown in Fig. 10 for high speed tool steel and Fe-Ni alloy. The powders are relatively spherical shaped. Table 1 shows fine metal powders that are called 'Fine Atomel'. These powders have recently been used in metal injection mouldings. However, other types of fine metallic powders can also be produced by this process.

CONCLUSIONS

Using two types of water jet nozzle (Cone and V-jet nozzle), the atomization mechanism



High speed tool steel



Fe-Ni alloy

FIG. 10 SEM micrographs of alloy powders produced by the high- pressured water atomization process.

Materials	Chemical Compositions (wt%)											Remarks
	C	Si	Mn	Ni	Cr	Mo	W	V	Co	Al	Fe	
4600	0.5/0.8	<0.20	0.10/0.35	1.7/2.2								Bal. AISI 4600
Stainless Steel 304L	<0.03	<1.0	<2.0	9.0/13.0	18.0/20.0							Bal. AISI 304L
Stainless Steel 410L	<0.03	<1.0	<2.0	<0.5	11.0/13.5							Bal. NF Z3C14
Stainless Steel 440C	0.95/1.20	<1.0	<1.0	<0.6	16.0/18.0							Bal. AISI 440C
Fe-Ni (46)	<0.02	<0.01	<0.01	45.5/46.5								Bal.
Fe-Ni (78)	<0.02	<0.01	0.01	77.5/78.5								Bal.
Fe-Co (30)	<0.02	<0.01	<0.01						29.5/30.5			Bal.
Kovar	<0.06		<0.5	28.0/30.0					16.0/18.5			Bal.
Fe-Si-Al (Sendust)	<0.03	9.5/10.5	<0.01							5.5/6.5		Bal.
Fe-Si (6)	<0.03	5.5/6.5	<0.01									Bal.
High Speed Tool Steel	1.20/1.35	<0.4	<0.4		3.8/4.5	3.0/4.0	9.0/11.0	3.0/3.7	9.0/11.0			Bal. Similar to AISI M36
Alloy Tool Steel	1.40/1.60	<0.4	<0.6		11.0/13.0	0.8/1.2		0.2/0.5				Bal. AISI D2

TABLE 1 Specification of 'Fine Atomel' for injection moulding.

under high pressure up to 70MPa and the properties of the atomized powders were investigated.

- (1) The V-jet nozzle has a higher pulverizing capability than the Cone nozzle.
- (2) The pulverizing capability greatly depends on the primary atomization of the inflow gas stream induced by the ejecting effects of water jets.
- (3) The powders produced through this process provide flow properties equivalent to

those of the powders produced by the carbonyl process.

REFERENCES

- (1) H Sakuma et al, 'Effect of Alloying Additions on the Properties of Partially Prealloyed Powders', Modern Developments in Powder Metallurgy, 19 (1988), 535.
- (2) T Takeda et al, Autumn Meeting of Japan Society of Powder and Powder Metallurgy, (1982), 16.